Interpretable Anomaly Detection – How explainability helps detect malicious network traffic, like cyber attacks.

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February 2026

1 Introduction

In the current zeitgeist, the field of machine learning is advancing rapidly, while the understanding of how such models work (XAI) is lagging behind (Dieber & Kirrane, 2020).

To investigate the importance of explainable AI methods, I will use a Random Forest classifier, which will be trained using the dataset CIC-IDS- 2017 (Sharafaldin et al., 2018) to classify normal and abnormal network activity. I have decided to experiment with different XAI methods and explore how it can benefit the field of cybersecurity. Today, network attacks are increasingly frequent. AI can be used in improving security by automating threat detection to prevent data privacy breaches and network attacks Annie Badman, 2026. This is done by establishing what normal network activity looks like, and any activity outside of this baseline is classified as a potential threat. However, the AI models used for such tasks can be biased. Not to mention the black box design, which is hard to interpret and understand, does not help the cybersecurity experts. Therefore, it is crucial to not overlook the explainability aspect of AI models.

2 Research Question

False negatives and False positives come at a high cost when it concerns AI use in Cybersecurity. Data breaches cost millions of USD annually. To minimize inaccurate labeling, to ensure that the system does not include any overlooked biases, and to minimize costs, explainability is a handy tool. XAI does not directly improve accuracy, but it helps cybersecurity experts to understand how specific decision was reached, and therefore help them identify weaknesses such as overfitting. I will explore what role XAI can play in anomaly detection in the domain of cybersecurity by questioning:

Do post-hoc explanations like LIME and DiCE can provide reliable and interpretable explanations when detecting anomalies in network data? Would counterfactual explanations (DiCE) provide more realistic measures of network traffic which cybersecurity experts can use to deal with network anomalies than feature attribution method like LIME?

3 Dataset

CIC-IDS-2017 is a dataset meant for intrusion detection. The data was being captured for 5 days in 2017. It covers variety of operating systems, most common protocols,

contains around 80 features. In this project, I only use the data that was collected on Friday, which consists of 3 total csv files.

4 Methods

I have chosen to explore the post-hoc explanation methods. Within this type, I picked LIME and Counterfactual Explanation method DiCE. Both of these methods are model-agnostic explanations. Comparing these two different methods within the same type of explanation allows me to contrast them and see how they explain the same cyber-attack.

Regarding the usefulness of the XAI methods, I will interpret the methods in therms of their faithfulness and human-interpretability measures. I will do so by playing the role of a security analyst: I will analyze how one could use the XAI results to interpret the network situation.

4.1 LIME- Local Interpretable Model-agnostic Explanation

LIME framework produces explanations, which provide insights into what role each feature plays for a specific sample. It functions by outputting probabilities of each class prediction and by indicating reliability of each explanation. LIME is Local, meaning that it focuses on one instance at a time. It can be useful for cybersecurity: for example, if a specific instance of network activity is labeled as malicious, LIME can offer insight, into why this was the case, therefore, giving an insight into how the security inference was made. This not only makes the system more transparent, but also more trustworthy and robust. Interpretable, means that it is easily understandable by humans, in this case we can interpret the features which the model considered. Model-agnostic means that it works for any black-box model, therefore it is robust and widely applicable (Dieber & Kirrane, 2020). Combination of such algorithm with AI model will allow to mitigate false labeling in real time, which can be costly in the field of cybersecurity. (Srivastava et al., 2022). LIME can unintentionally expose undesired information from dataset, for example, by highlighting specific features which were used in decision making, like IP addresses (Ofusori et al., 2025)

4.2 DiCE- Diverse Counterfactual Explanations

DiCE functions in a "what if" manner: given the output and input, what would the output be if the input changed from x to x'. Therefore, Counterfactual explanations are designed to identify what minimum change would also change the classification result (Pawlicki et al., 2024b).

Since DiCE functions by creating alternative situations, which result in opposite labeling, it could be used not only to combat cyber-attacks, but also to craft attacks that can evade classification algorithms. Overall, DiCE not only explains how decisions were made, it also shows how they can be potentially manipulated.

The equation (1) works by finding minimal change that flip the classification result:

$$\min_{\{c_1,\ldots,c_k\}} (\frac{1}{k}\sum_{i=1}^{k}[\lambda_1 \cdot \ell(f(c_i), y') + \text{dist}(c_i, x)] - \lambda_2 \cdot \text{dpp_diversity}(c_1, \ldots, c_k)) \quad (1)$$

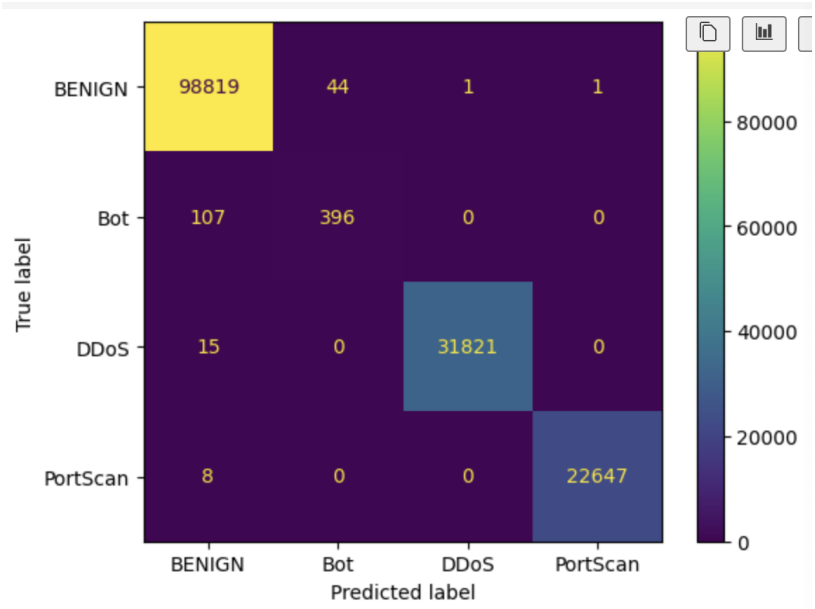
DiCE differs from LIME, since they ask and answer different questions. LIME asks "What lead to this prediction?", while DiCE questions "What has to change for the prediction to change?". Both of the mentioned methods are model-agnostic, meaning they are applicable to any black-box model.

5 Programming Part

I loaded the dataset from (CIC-IDS2017). I chose to use only the files gathered during Friday. I loaded the .csv files, merged them into a single data frame. The result was a data frame shaped (703245, 79). Then I preprocessed data by removing missing values, fixing formatting issues, creating train-test split. For training the model on given data, I used the Random Forest Classifier form scikit learn library in python ((“RandomForestClassifier,” n.d.)).

I evaluated the performance to see how well I performed in the previous step, by evaluating the Accuracy and Recall, F1 score measures, plotted the confusion matrix. Despite observing high accuracy measures, the confusion matrix showed that ”Bot” class is not being labeled correctly by my model. This is likely due to the fact that ”Bot” label counts are highly imbalanced compared to other classes.

	Label	count
0	BENIGN	394777
1	DDoS	128014
2	PortScan	90694
3	Bot	1948



Further explanations of code and results of measures are provided in the Jupyter notebook located at the GitHub:

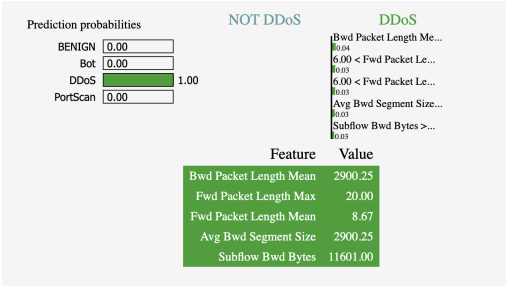
https://github.com/rasavaiciulyte/XAI_final_submission.git

5.1 Applying XAI methods

After training, I have started to tackle implementation of XAI methods.

5.1.1 LIME

I applied LIME explanations for individual label classes and showcased the explanation output for at least one instance of each class.



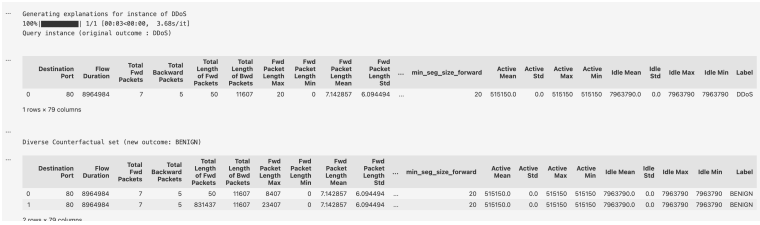
The results were not as expected. In this instance the model is very confident in the predicted label, features in the explanation all are in favor of "DDoS" with no countering feature values. However, the feature importance seem to be very low.

The model seems very confident in most labels (except for "Bot" due to class imbalance). Plotted feature importance seems to not explain the choice that well. I suspected that this might be due to overfitting and mistakes when dealing with data or, perhaps, due to the fact that there is no single feature that sets off a specific labeling outcome, but rather a combination of small signals all together.

This seems not as interpretable and as I expected and, I can only speculate, that does not seem like a faithful explanation which cybersecurity experts could use with high confidence. For further evaluation of the method, it would be beneficial to ask a domain expert whether they agree with such motivations of explanation.

5.1.2 DICE

To gain insights into how DiCE works in practice I used the following documentation website (https://interpret.ml/DiCE/notebooks/DiCE_getting_started.html)



```
Generating explanations for instance of DDoS
100%|▇▇▇▇▇▇▇▇▇▇| 1/1 [00:03<00:00,  3.68s/it]
Query instance (original outcome : DDoS)
```

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets	Total Length of Bwd Packets	Fwd Packet Length Max	Fwd Packet Length Min	Fwd Packet Length Mean	Fwd Packet Length Std	...	min_seg_size_forward	Active Mean	Active Std	Active Max	Active Min	Idle Mean	Idle Std	Idle Max	Idle Min	Label
0	80	8964984	7	5	50	11607	20	0	7.142857	6.094494	...	20	515150.0	0.0	515150	515150	7963790.0	0.0	7963790	7963790	DDoS

1 rows × 79 columns

```
Diverse Counterfactual set (new outcome: BENIGN)
```

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets	Total Length of Bwd Packets	Fwd Packet Length Max	Fwd Packet Length Min	Fwd Packet Length Mean	Fwd Packet Length Std	...	min_seg_size_forward	Active Mean	Active Std	Active Max	Active Min	Idle Mean	Idle Std	Idle Max	Idle Min	Label
0	80	8964984	7	5	50	11607	8407	0	7.142857	6.094494	...	20	515150.0	0.0	515150	515150	7963790.0	0.0	7963790	7963790	BENIGN
1	80	8964984	7	5	831437	11607	23407	0	7.142857	6.094494	...	20	515150.0	0.0	515150	515150	7963790.0	0.0	7963790	7963790	BENIGN

2 rows × 79 columns

For DICE I choose to focus on 10 features of the dataset, so the explanation could work properly. This explanation seems more trustworthy and interpretable than LIME. It provides useful insights into what can be changed for the network activity to be labeled as unharmful.

6 Results

Resulting plots of LIME explanations show what features and their values lead to such labeling decisions. However, their influence is very close to 0, suggesting that decision making is based on a combination of various factors. This can be due to the fact that the dataset contains 80 features, which corresponds to a lot of dimensions. It seems very counter intuitive, that decisions are made with high confidence, but not much influence of specific features individually. The insights generated by DiCE felt more actionable than lime in showcasing the boundary between a "DDoS" and "BENIGN" labels, which can cost a lot of money if mislabeled in a real-world situation.

7 Discussion

The paper by Pawlicki et al., 2024a takes this critical approach on CE (Counterfactual Explanation) methods. The authors question, whether algorithms, such as DiCE can be used to formulate attacks on AI based abnormal network activity detection. XAI methods are compared to a dual-edged sword, due to the fact that, the same mechanisms that make AI explainable can be used against us (Capuano et al., 2022).

8 Conclusions

Overall, post-hoc methods like LIME and DiCE when used on labeling network anomalies, provide meaningful insights when compared side by side. This is due to the fact that they function by different principals: while one focuses on what lead to the decision, the second one gives insights into what could be changed to flip the label result.

All in all, XAI methods reveal more about the model than the accuracy measures. The use of explainability is beneficial in many fields, and cybersecurity is one of them. XAI can be used to improve readability and aid in model improvements. However, we it is crucial to consider the pitfalls of such methods. We should not overlook the mentioned possibility of XAI being used not only by defenders of traffic, but also by its attackers. Lastly, we should be careful not to develop a false sense of security that XAI may create.

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